

$$A = \begin{bmatrix} | & | & | & | & | \\ p_1 & f_1 & p_2 & p_3 & f_2 \\ | & | & | & | & | \end{bmatrix}$$

$$P = \begin{bmatrix} | & | & | \\ p_1 & p_2 & p_3 \\ | & | & | \end{bmatrix}$$

no col is a lin. comb. of any other col.

$\text{Col}(A) =$ set of all **linear comb** of cols of A .
 $= \{ Ax : x \in \mathbb{R}^5 \}$.

$$\begin{bmatrix} | & | & | & | & | & | \\ p_1 & f_1 & p_2 & p_3 & f_2 & Ax \\ | & | & | & | & | & | \end{bmatrix}$$

$$\begin{bmatrix} | & | & | & | \\ p_1 & p_2 & p_3 & Ax \\ | & | & | & | \end{bmatrix}$$

$$Ax = PCx = Py \quad (\text{where } y = Cx)$$

$$A = \begin{bmatrix} | & | & | \\ p_1 & p_2 & p_3 \\ | & | & | \end{bmatrix} \begin{bmatrix} | & 0 & 0 \\ 0 & \text{second} & 0 \\ 0 & \text{col of} & 1 \end{bmatrix} \begin{bmatrix} | & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A = PC \quad (C = \text{first 3 rows of } \text{rref}(A)).$$

All max linear independent sets in a linear space L have the same size.

Proof by contradiction.

Suppose $V = \{v_1, \dots, v_n\}$ and $W = \{w_1, \dots, w_m\}$ are both max linearly indpt. Without loss of generality assume $n > m$.

$$W' = \begin{bmatrix} | & & | & | \\ \frac{1}{w_1} & \dots & \frac{1}{w_m} & \frac{1}{v_1} \\ | & & | & | \end{bmatrix}$$

$$V = W \begin{bmatrix} \alpha_{11} & \dots & \alpha_{1n} \\ \alpha_{21} & \dots & \alpha_{2n} \\ \vdots & & \vdots \\ \alpha_{m1} & \dots & \alpha_{mn} \end{bmatrix}$$

Some col must be free, since $n > m$.

$$\beta_1 c_1 + \beta_2 c_2 + \dots + \beta_{k-1} c_{k-1} - c_k = 0$$

$$c_k = \text{linear comb of } c_1 \dots c_{k-1} \stackrel{\text{say}}{=} \beta_1 c_1 + \beta_2 c_2 + \dots + \beta_{k-1} c_{k-1}$$

$$\therefore \underbrace{\begin{bmatrix} \alpha_{11} & \dots & \alpha_{1n} \\ \alpha_{21} & & \alpha_{2n} \\ \vdots & & \vdots \\ \alpha_{m1} & & \alpha_{mn} \end{bmatrix}}_W \cdot \begin{pmatrix} f_1 \\ f_2 \\ \vdots \\ f_{k-1} \\ 0 \\ \vdots \end{pmatrix} \stackrel{\vec{z}}{=} \underline{0}.$$

$$V\vec{z} = W \underbrace{\begin{bmatrix} \alpha_{11} & \dots & \alpha_{1n} \\ \alpha_{21} & & \alpha_{2n} \\ \vdots & & \vdots \\ \alpha_{m1} & & \alpha_{mn} \end{bmatrix}}_W \vec{z}$$

Null space of A

$$\text{null}(A) = \{ \vec{z} : A\vec{z} = \underline{0} \}.$$

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